

Autonomous and Domain-Specific Agentic AI Data Workforce for Indian Enterprises

Developed at IIT Madras | Researched in Harvard Dataverse



Indian enterprises manage vast volumes of sensitive operational, financial, and customer data, traditionally processed through manual spreadsheets or fragmented BI tools. While Large Language Models (LLMs) offer revolutionary potential for data intelligence, traditional cloud-based AI deployments require the transfer of raw data to third-party servers. This creates a critical conflict with the Digital Personal Data Protection (DPDP) Act and poses significant risks to corporate sovereignty and proprietary intelligence. Enterprises must find a way to leverage agentic AI to eliminate technical debt and manual bottlenecks, but in a way that ensures data never leaves the organization's perimeter.

YData, developed at IIT Madras, resolves this conflict by deploying a sovereign, autonomous AI data workforce designed specifically for the Indian regulatory and business landscape. Built on a proprietary framework published at Harvard Dataverse, YData utilizes a sophisticated local-first engine to bring the AI "brain" to the data, rather than the data to the AI. Its agentic engine autonomously assesses datasets to identify forensic anomalies, map technical schemas to business language, and uncover strategic growth segments, all while executing 100% of the mathematical processing within the user's secure browser environment.

Key Benefits

Compared to traditional approaches, YData enables:

- ❖ **100%** data sovereignty
- ❖ **5x** faster auditing & analysis
- ❖ **10x** faster business intelligence
- ❖ **100%** Bharat-Ready
- ❖ **Zero** technical requirement
- ❖ **100%** automated AI workforce
- ❖ **Zero** compute costs

Algorithmically benchmarked
against data from



भारतीय रिज़र्व बैंक
RESERVE BANK OF INDIA

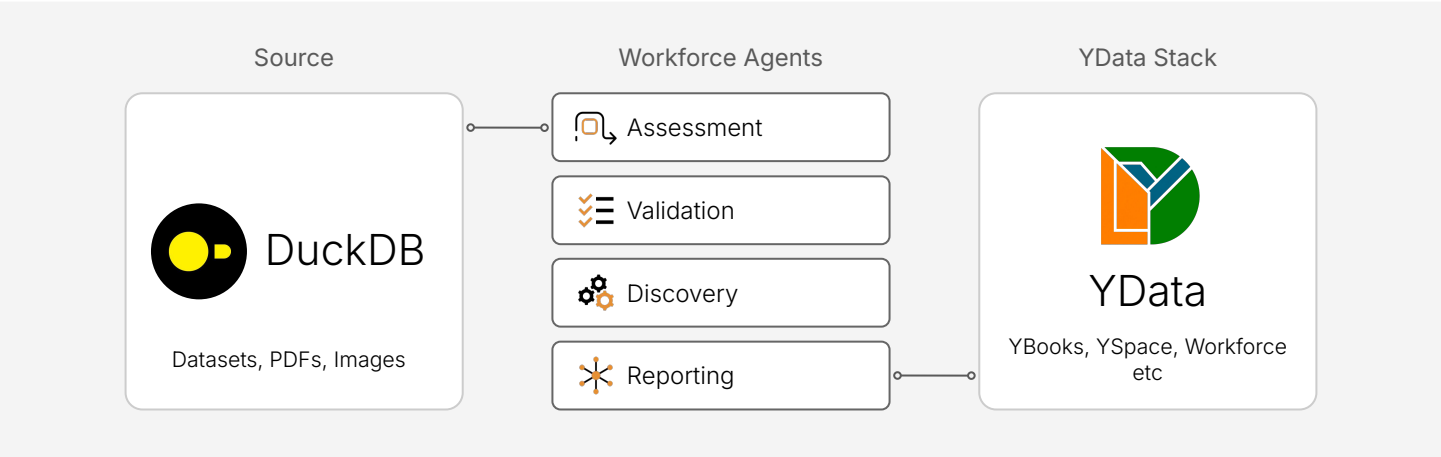


data.gov.in
Open Government Data (OGD) Platform India



Key Benefits

YData automates the end-to-end data lifecycle through four distinct agentic phases, transforming raw files into boardroom-ready intelligence.

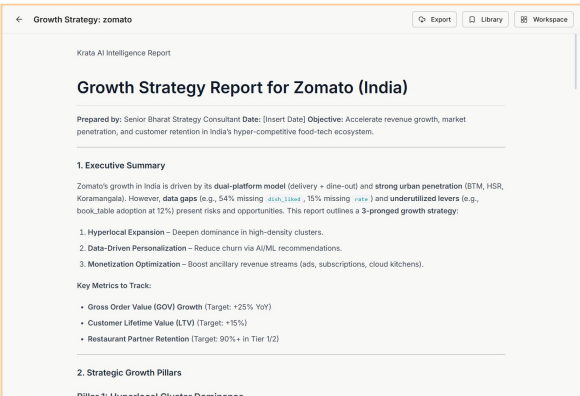
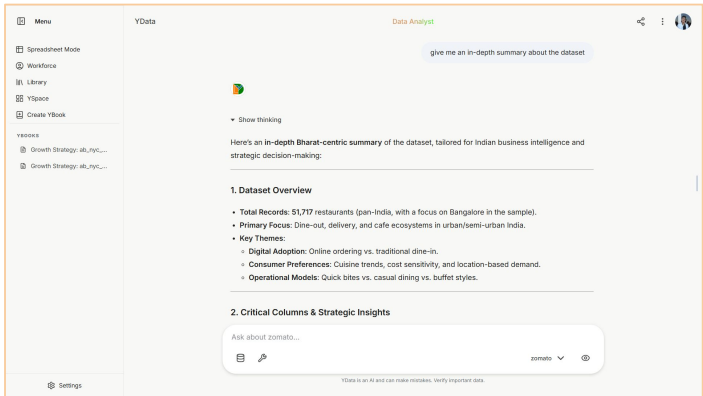


STEP 1: Ingestion & Mapping

- ❖ **Multimodal Processing:** Utilizing local-first OCR to ingest unstructured PDFs and images alongside standard structured files (CSV, Excel, JSON).
- ❖ **Semantic Translation:** Automatically identifies cryptic technical headers and utilizes LLM-based type inference to map them into business language (e.g., VAR_99 → Net Revenue).
- ❖ **Privacy-First Indexing:** Data is indexed into an in-browser DuckDB-Wasm instance. Raw data never touches YData servers.

STEP 2: Integrity & Governance

- ❖ **PII Shield:** Automated detection of sensitive PII (Emails, Indian Phone Numbers, Credit Cards, Gov IDs) with one-click local redaction to ensure DPDP Act compliance.
- ❖ **Statistical Cleansing:** Autonomous identification of sparsity, null values, and type-mismatches with statistical imputation to ensure high-fidelity analysis.
- ❖ **Forensic Audit:** Real-time advanced analyses to detect manual data manipulation, rounding errors, or potential financial fraud in ledger data.



STEP 3: Strategic Discovery

- ❖ **Pareto Analysis (80/20 Rule):** The engine autonomously calculates the Gini coefficient to identify "Whale" segments - the 20% of drivers generating 80% of your business impact.
- ❖ **Bharat-Localized Intelligence:** Deep analysis of Indian business cycles, including GST slab reconciliation, Fiscal Year (Apr-Mar) trends, and festive seasonality.
- ❖ **Trend Spotting:** Agentic identification of statistical outliers and growth pockets in Tier-2 and Tier-3 Indian cities.

STEP 4: Autonomous Reporting

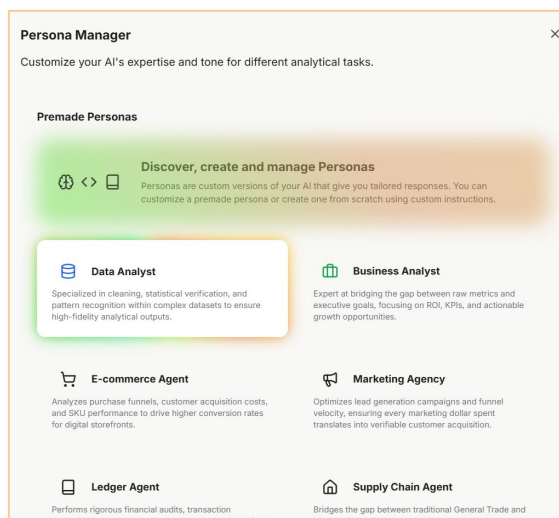
- ❖ **Executive Synthesis:** Instant generation of KrataBooks - high-fidelity BI reports that translate complex findings into strategic executive narratives.
- ❖ **Visual Intelligence:** Automated creation of 12 archetypes of visualizations (Funnels, User Journeys, Gantt Charts) embedded directly in reports.
- ❖ **Universal Export:** Professional PDF and CSV exports for seamless sharing with stakeholders and regulatory bodies.

Modular Persona Intelligence

YData features an extensible Agentic Layer, allowing enterprises to deploy specialized virtual experts tailored to their specific industry, department, or regulatory requirements.

Domain-Adaptive Intelligence:

Unlike generic AI models, the YData engine allows users to toggle between diverse analytical "brains." Each persona is engineered with proprietary logic to interpret data through the lens of specific business functions - from high-stakes forensic auditing to last-mile supply chain optimization.



To automate your data workflow using agentic AI, write to the founding team at
22f3000795@ds.study.iitm.ac.in



Licensing and Investment Tiers

Join a limited cohort of enterprises deploying IIT developed agentic intelligence to automate data forensics and strategic reporting.

The 3 Tiers of Intelligence

1. STANDARD (The Departmental License)

Ideal for small boutique agencies or specific departments within a larger firm.

- ❖ **Investment: ₹1.8 Lakhs / Year (10% Discounted from ₹2L)**
- ❖ Capacity: Up to 5 User Seats (₹36k per user)

2. PROFESSIONAL (The Business License) — [MOST POPULAR]

Designed for mid-sized Indian enterprises and high-growth D2C brands.

- ❖ **Investment: ₹8 Lakhs / Year (33% Discounted from ₹12L)**
- ❖ Capacity: Up to 35 User Seats (₹23k per user)

3. ENTERPRISE (The Site License)

For large-scale corporations requiring custom integration and absolute scale.

- ❖ **Investment: ₹35 Lakhs / Year (Custom pricing scales with additional features)**
- ❖ Capacity: 100 Users to Unlimited Users Site License
- ❖ Exclusive Features:
 1. Custom Persona Training: We build virtual experts specifically for your proprietary ERP/Data structures.
 2. SSO & Private Instance: Integrated Okta/Azure AD login and dedicated Supabase infrastructure.
 3. White-Labeled Application: Custom-branded Application with it's own domain for your enterprise.
 4. 24/7 Dedicated Support: Direct WhatsApp/Phone line to the engineering team.

